
Averaged Jacobian Regularity Can Misrank Few-Step Flow-Matching Schedules: A Certified Gaussian Counterexample

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Abstract

1 Few-step sampling in flow matching and stochastic interpolants requires choosing
2 an interpolation schedule before endpoint error is observed. Averaged squared
3 Lipschitzness of the drift, measured by the time-integrated squared 2-norm of its
4 spatial Jacobian, is a plausible schedule-design criterion, but it is not a solver-
5 specific discretization-error functional. Chen, Vanden-Eijnden, and Xu propose
6 minimizing this quantity, denoted A_2 ; they do not claim or prove that A_2 univer-
7 sally ranks equal-NFE solver error. We study that narrower surrogate question. For
8 independent $N(0, 1)$ and $N(0, 4)$, the exact regularity integrals are $5\pi/8 - 1$ and
9 $\pi^2/16$, so regularity prefers trigonometric VP, whereas explicit Heun, a two-stage
10 Runge–Kutta method, with a budget of eight function evaluations (NFE 8), prefers
11 the linear path in Gaussian Wasserstein-2 distance. The inversion is certified by
12 an exact rational Heun product and a nonnegative element of $\mathbb{Q}[\pi, \sqrt{2}]$. A com-
13plementary construction shows that, for every step count N and every admissible
14 endpoint log-scale, the unique integrated-regularity minimizer can have strictly
15 larger N -step Euler endpoint error than a higher-regularity competitor aligned with
16 that solver grid. A specified finite Gaussian enumeration of four candidate paths on
17 36 tested blocks contains both agreement and pairwise disagreement. These results
18 limit a universal surrogate interpretation; they do not evaluate learned velocity
19 fields or estimate a population failure frequency.

20 1 Introduction

21 Few-step sampling commits to a path, a solver, and a small NFE budget before any error is observed.
22 Lipschitz constants enter classical one-step bounds [Hairer et al., 1993]. Chen et al. [2025] propose
23 minimizing A_2 , the averaged squared Lipschitzness of the drift (their Definition 3.2), as a schedule-
24 design objective in a specified admissible family. They do not claim or prove that A_2 universally
25 ranks equal-NFE solver error. We study that narrower surrogate question on a flow-matching marginal
26 ODE.

27 The object is an integrated unsigned Jacobian energy. For a fixed finite-step solver, endpoint error
28 depends on stagewise field values, temporal derivatives, quadrature structure, and signed cancellations
29 that this energy does not determine. For independent $N(0, 1)$ and $N(0, 4)$, exact regularity prefers
30 trigonometric VP while Heun at NFE 8 prefers the linear path in Gaussian W_2 (Section 3). A
31 solver-grid-aware construction then shows that even the global $\mathcal{R}$ -minimizer need not minimize
32 fixed-grid Euler error (Section 6); the competitor is built from the solver resolution. A finite Gaussian
33 enumeration of four specified candidate paths on 36 tested blocks contains both agreement and
34 pairwise disagreement (Section 5). No learned field is used.

Venue relevance is through distribution learning, Wasserstein endpoint error, and the geometry of Gaussian covariance interpolation, not through a new geometric architecture or a practical OT algorithm. Tao and Choi [2026] give a broader dynamical error analysis, including material derivatives, strain, vorticity, logarithmic norms, and learned models. The present contribution is narrower but exact and auditable: a certified scalar Gaussian ranking inversion, a closed-form W_2 comparison, a grid-aware impossibility construction, and a transparent finite enumeration. Contemporaneous Lipschitz Euler rates [Stéphanovitch, 2026] bound a different object.

2 Gaussian interpolants and averaged regularity

Let $X_0 \sim N(\mu_0, \Sigma_0)$ and $X_1 \sim N(\mu_1, \Sigma_1)$ be independent. A shared schedule is a scalar pair (α, σ) . We distinguish three regularity classes.

- Class S: $\alpha, \sigma \in C^1([0, 1])$ with $\alpha(0) = \sigma(1) = 1$, $\alpha(1) = \sigma(0) = 0$, and $(\alpha(t), \sigma(t)) \neq (0, 0)$ on $[0, 1]$. Linear and trigonometric VP belong to Class S.
- Class Q: a C^1 path of SPD covariances $Q(t)$ with continuous $A = \frac{1}{2}Q'Q^{-1}$. The per-mode schedule $q_i(t) = \lambda_i^t$ is Class Q.
- Class R: $\alpha^2, \sigma^2 \in C^1([0, 1])$, $\alpha^2 \geq 0$, $\sigma^2 \geq 0$, $\alpha^2 + \sigma^2 > 0$, and $\alpha(0)^2 = 1$, $\sigma(0)^2 = 0$, $\alpha(1)^2 = 0$, $\sigma(1)^2 = 1$. Square roots need not be C^1 at the endpoints. Chen Ex. 3.3, $\sigma(t)^2 = (M^t - 1)/(M - 1)$ and $\alpha(t)^2 = (M - M^t)/(M - 1)$ for $M > 0$, $M \neq 1$, is Class R and Class Q, not Class S. The $M = 1$ limit is $\sigma(t)^2 = t$, $\alpha(t)^2 = 1 - t$.

The interpolant $X_t = \alpha X_0 + \sigma X_1$ [Albergo et al., 2025, Lipman et al., 2023] is Gaussian with mean $m_t = \alpha\mu_0 + \sigma\mu_1$ and covariance $Q = \alpha^2\Sigma_0 + \sigma^2\Sigma_1$. The marginal field is affine,

$$b(t, x) = A(t)x + c(t), \quad A = C_t Q^{-1}, \quad c = \dot{m} - Am. \quad (1)$$

The formula $C_t = \frac{1}{2}(\alpha^2)' \Sigma_0 + \frac{1}{2}(\sigma^2)' \Sigma_1$ extends continuously to $[0, 1]$ for Class R; equivalently $C_t = \dot{\alpha}\alpha \Sigma_0 + \dot{\sigma}\sigma \Sigma_1$ on $(0, 1)$ where the coefficients are differentiable and nonzero. This is a flow-matching marginal ODE, not a score-based probability-flow ODE in the sense of Song et al. [2021]. If $\Sigma_0 = I$, then Q , Q' , A , and Σ_1 share the eigenbasis of Σ_1 , and mode i obeys $x'_i = a_i(t)x_i + c_i(t)$ with $q_i = \alpha^2 + \lambda_i\sigma^2$ and $a_i = q'_i/(2q_i)$. The covariance drift formula is independent of the mean path $c(t)$. We compare the linear path $(1 - t, t)$ and trigonometric VP $(\cos(\pi t/2), \sin(\pi t/2))$. Following Chen et al. [2025, Definition 3.2], $\mathcal{R}[b] = \int_0^1 \|A(t)\|_2^2 dt$. For state-independent A this coincides with their path expectation. By definition $\mathcal{R}$ ignores $c(t)$. Error is Gaussian W_2 of the endpoint laws [Gelbrich, 1990]. This is neither pathwise trajectory error nor local truncation error. Equal NFE uses $S = B/s$ steps with $s = 1, 2, 4$ for Euler, Heun, and RK4.

Any regular variance path from 1 to $\lambda > 0$ obeys $\int_0^1 a dt = \frac{1}{2} \ln \lambda$. Hence $\mathcal{R} = \int_0^1 a^2 dt = (\int a)^2 + \text{Var}_t(a) \geq (\frac{1}{2} \ln \lambda)^2$, with equality if and only if a is constant. The one-dimensional minimizer is $q(t) = \lambda^t$, $a \equiv \frac{1}{2} \ln \lambda$, which is Chen et al. Example 3.3. Linear and trigonometric VP are C^1 shared interpolants. In several dimensions, the per-mode schedule $q_i(t) = \lambda_i^t$ yields $\mathcal{R} = (\frac{1}{2} \max_i |\ln \lambda_i|)^2$. When Σ_1 has at least two distinct eigenvalues, that path is not a Class S shared interpolant.

3 A one-dimensional Gaussian ranking counterexample

Assumption 1. Endpoints $N(0, 1)$ and $N(0, 4)$ are independent. Schedules are linear and trigonometric VP. The solver is explicit Heun with NFE 8 (four steps of size $1/4$).

Proposition 1. Under the assumption, $\mathcal{R}_{\text{lin}} = 5\pi/8 - 1 > \mathcal{R}_{\text{VP}} = \pi^2/16$, while the Heun factors satisfy $r_{\text{lin}} = 6797469/3559400$ and $0 < r_{\text{VP}} < 187/100$. Hence $W_2^{\text{lin}} < 0.091 < 0.130 < W_2^{\text{VP}}$.

Proof sketch. The linear variance is $q_{\text{lin}}(t) = 1 - 2t + 5t^2$. Completing the square gives the equivalent identities $q_{\text{lin}}(t) = 5[(t - 1/5)^2 + (2/5)^2] = 5(t - 1/5)^2 + 4/5$. The substitution $u = t - 1/5 = (2/5) \tan \theta$ converts $\int a_{\text{lin}}^2$ into $(5/2) \int \sin^2 \theta d\theta$. Limits $\arctan(-1/2)$ and $\arctan 2$ together with $\arctan 2 + \arctan(1/2) = \pi/2$ yield $\mathcal{R}_{\text{lin}} = 5\pi/8 - 1$. The VP integral evaluates to $\pi^2/16$ by a Weierstrass identity. The comparison $5\pi/8 - 1 > \pi^2/16$ rearranges to $(\pi - 2)(\pi - 8) < 0$,

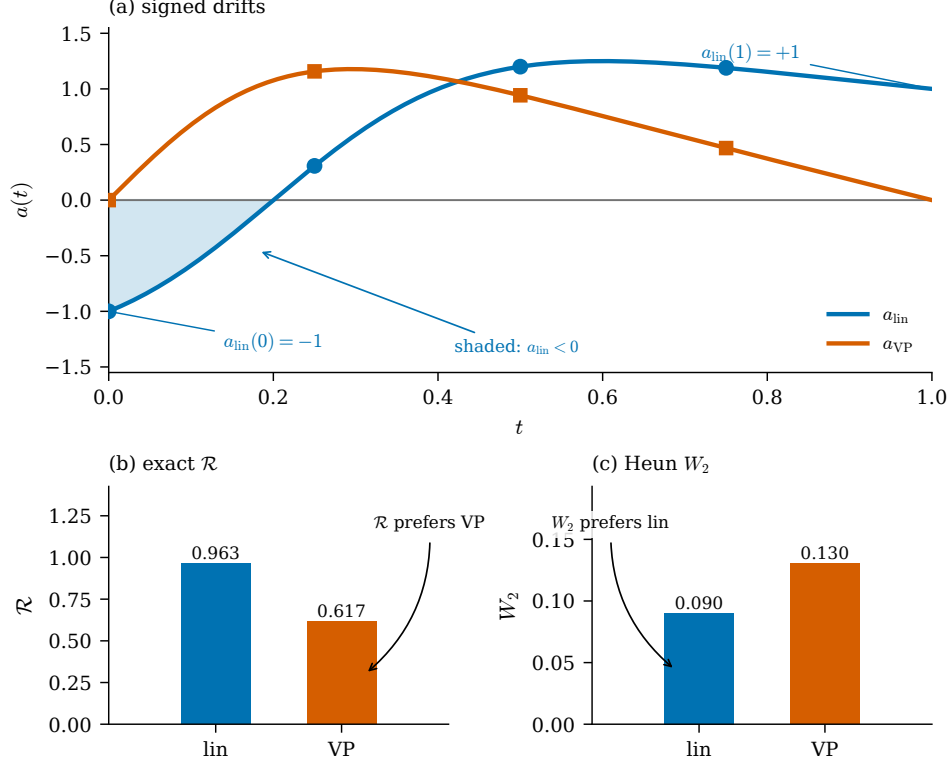


Figure 1: Certified scalar counterexample for independent $N(0, 1)$ and $N(0, 4)$: drifts, exact $\mathcal{R}$ integrals, and Heun NFE 8 Gaussian W_2 .

81 which holds because $2 < \pi < 8$. The four linear Heun factors are the rationals $47/52$, $6/5$, $497/370$,
82 and $97/74$, all positive. Exact VP grid values are $a(0) = a(1) = 0$, $a(1/2) = 3\pi/10$, and $a(1/4) =$
83 $(9 + 15\sqrt{2})\pi/82$, $a(3/4) = (-9 + 15\sqrt{2})\pi/82$. Positivity of $a(3/4)$ is $15^2 \cdot 2 - 9^2 = 369 > 0$. The
84 four-step product lies in $\mathbb{Q}[\pi, \sqrt{2}]$ with nonnegative coefficients, so substituting $\sqrt{2} < 99/70$ and
85 $\pi < 355/113$ is valid and yields $r_{VP} < 187/100$ by an integer comparison in Appendix A. Interval
86 arithmetic is a cross-check only. $\square$

87 4 Sign change and transported defects

88 For any independent linear coupling, $a_{lin}(0) = -1 < 0 < a_{lin}(1) = +1$. The VP drift vanishes at
89 the endpoints and does not change sign on $(0, 1)$. Averaging a^2 does not retain that sign change.
90 Fixed-stage Runge–Kutta methods use signed stage evaluations that are not determined by $\mathcal{R}$. The
91 sign difference is consistent with the observed inversion, but the analysis does not isolate it as a causal
92 contribution.

93 Let $e_n = \sqrt{q(t_{n+1})/q(t_n)}$ and let r_n be the numerical factor. The telescoping identity

$$\prod_n r_n - \prod_n e_n = \sum_n (r_n - e_n) \left(\prod_{j < n} r_j \right) \left(\prod_{j > n} e_j \right) \quad (2)$$

94 is an accounting tool, not the content of the inversion. The content is the leading exact-minus-method
95 coefficient: $\frac{1}{2}(a' + a^2)h^2$ for Euler and $(-a''/12 + a^3/6)h^3$ for the implemented Heun factor. Those
96 expansions involve different derivatives, signs, and stage locations.

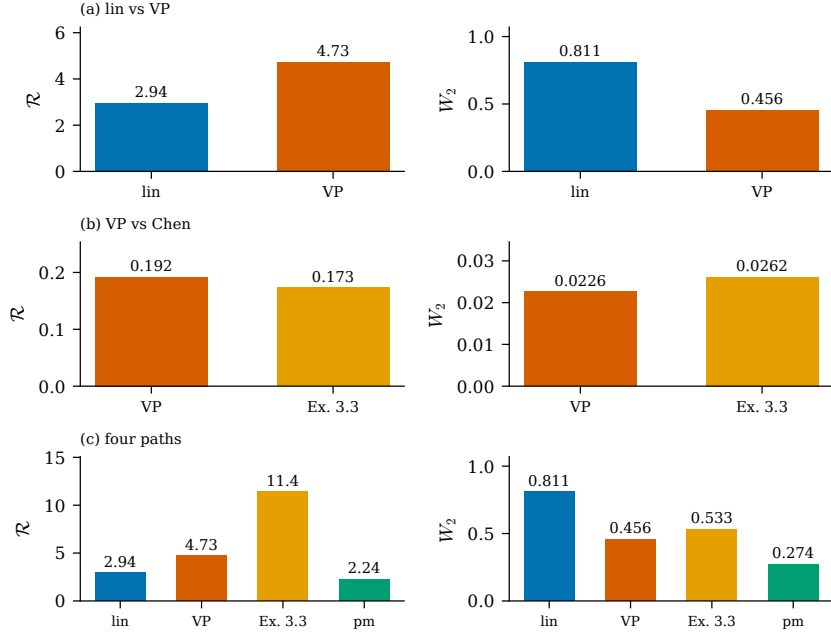


Figure 2: Three comparisons. Left: $\mathcal{R}$. Right: Euler NFE 8 W_2 . (a) linear vs VP, rank-2 factor-plus-noise $d = 8$; (b) VP vs Chen Ex. 3.3, anisotropic $d = 8$; (c) four specified paths, rank-2 factor-plus-noise $d = 8$. Among those four paths, per-mode wins $\mathcal{R}$ and W_2 in 36 of 36 tested blocks.

Table 1: Four specified schedules at Heun, NFE 8. Among these four candidate paths, per-mode has the smallest W_2 on every geometry in the 36 tested blocks. When Σ_1 has at least two distinct eigenvalues, the per-mode path is not a C^1 shared interpolant.

family	d	$\mathcal{R}_{\text{lin}}$	$\mathcal{R}_{\text{VP}}$	$\mathcal{R}_{\text{sc}}$	$\mathcal{R}_{\text{log}}$	Heun-8 W_2 min
anisotropic	2	0.972	0.192	0.173	0.120	log-cov
anisotropic	8	0.972	0.192	0.173	0.120	log-cov
factor+noise	2	1.246	0.625	0.668	0.380	log-cov
rank-2 factor+noise	8	2.944	4.731	11.419	2.244	log-cov

97 5 Four-path enumeration and the inversion region

98 Source $N(0, I)$. Four centered targets: anisotropic Gaussians with condition number 4 in dimensions
 99 2 and 8 (eigenvalues a geometric sequence from $1/2$ to 2); factor-plus-noise covariances $\Sigma =$
 100 $FF^\top + 0.05I$ with $F \in \mathbb{R}^{d \times 2}$ i.i.d. standard normal (seed 271828). When $d = 2$ the factor rank
 101 equals the ambient dimension. Because $\mathcal{R}$ uses the spectral norm, the two anisotropic dimensions
 102 share the same $\mathcal{R}$ pair (0.972296, 0.192270), so the grid contains three distinct linear-versus-VP $\mathcal{R}$
 103 comparisons. Each geometry is run under Euler, Heun, and RK4 at NFE $\{8, 16, 32\}$. Headline $\mathcal{R}$ is
 104 the continuous integral of $\max_i a_i(t)^2$. Ranking inversions were identified in Phase 3; the counts
 105 below are a confirmatory Phase 4 reproduction on configurations frozen before the rerun.

106 The per-mode log-covariance path $q_i(t) = \lambda_i^t$ is one of four specified candidate paths. Among those
 107 four paths on the stated 36 tested blocks it has the smallest $\mathcal{R}$ on every geometry and the smallest
 108 Gaussian W_2 in all 36 tested blocks (Table 1 reports Heun at NFE 8). This finite enumeration
 109 does not imply that a global $\mathcal{R}$ -minimizer minimizes fixed-NFE error. When Σ_1 has at least two
 110 distinct eigenvalues, that path is not a C^1 shared interpolant. Linear versus VP invert in 5 of 12
 111 geometry $\times$ solver cells; 4 of those 12 invert at every tested NFE. As a post-hoc pairwise comparison
 112 within the shared-schedule class, trigonometric VP versus Chen Ex. 3.3 with $M = \lambda_{\max}$ invert in 9
 113 of 36 tested blocks and 4 of 12 cells.

Table 2: Strongest centered inversion: rank-2 factor-plus-noise, $d = 8$, Euler, NFE 8. Eigenvalues: 0.05 (six times), 4.64827725629505, 10.675638226546441. Deterministic calculation: $\mathcal{R}$ is evaluated by adaptive quadrature, while W_2 uses exact Gaussian moments propagated through the modal Runge–Kutta updates.

path	$\mathcal{R}$	$\hat{\mathcal{R}}_{24}$	Gaussian W_2
linear	2.9441044083	2.9476523251	0.8108540111
VP	4.7305438136	4.7295206355	0.4564779075

114 A one-dimensional sweep $\lambda \in [0.05, 100]$ (Figure 3) locates the linear-versus-VP inversions. At $\lambda =$
 115 4, the inversion is present under Heun at every tested budget NFE $\in \{4, 8, 12, 16, 24, 32, 64, 128, 256\}$,
 116 absent under Euler, and mixed under RK4 (absent at NFE 8 and 12). Euler *does* invert at more
 117 extreme λ on the same sweep. The scalar example is therefore solver-dependent, not an artefact of
 118 Heun at budget 8 alone.

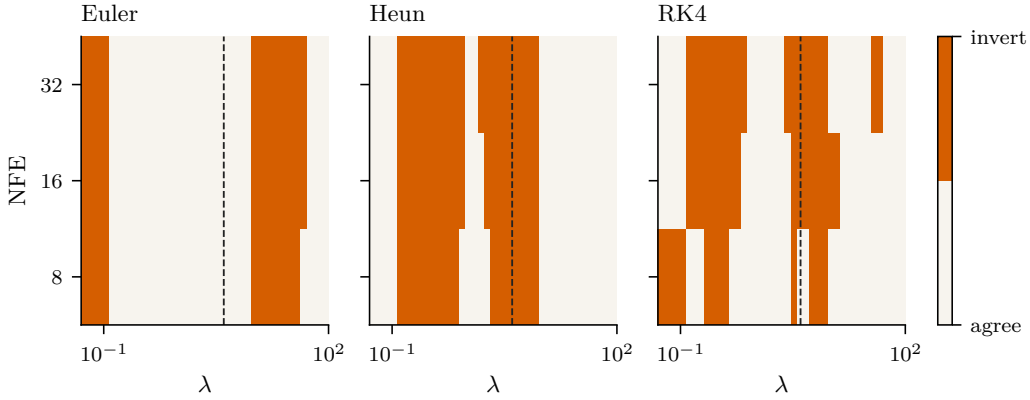


Figure 3: Linear versus VP ranking inversion versus target variance λ . Dark cells: $\mathcal{R}$ and W_2 disagree. The dashed line is $\lambda = 4$ (Proposition 1).

119 6 A grid-aware Class S Euler construction

120 **Theorem 1.** Fix $L > 0$ and an integer $N \geq 1$. Let $h = 1/N$ and let forward Euler use left endpoints
 121 $t_n = nh$ on $[0, 1]$. Set $\varepsilon_N = N(e^{L/N} - 1) - L > 0$, $a_0(t) \equiv L$, and $a_{1,N}(t) = L + \varepsilon_N \cos(2\pi Nt)$.
 122 For $x' = a(t)x$, $x(0) = 1$, and the Class S embeddings $q_k = \exp(2 \int_0^t a_k)$, $\alpha_k = \sqrt{q_k} \cos(\pi t/2)$,
 123 $\sigma_k = \sqrt{q_k}/e^{2L} \sin(\pi t/2)$: both exact endpoints equal e^L ; a_0 uniquely minimizes $\int a^2$ among scalar
 124 fields with integral L ; Euler is exact for $a_{1,N}$ and strictly inexact for a_0 . Endpoint error is $|r - e^L|$,
 125 equivalently Gaussian W_2 between $N(0, r^2)$ and $N(0, e^{2L})$.

126 The competitor is constructed as a function of the solver grid. Phase offsets and non-grid frequencies
 127 destroy exact Euler aliasing. This is a solver-grid-dependent impossibility result, not a model of
 128 natural schedules, and not the mechanism of Proposition 1.

129 7 Related work

130 Flow matching and interpolants supply the Gaussian family [Lipman et al., 2023, Albergo et al.,
 131 2025, Liu et al., 2023, Tong et al., 2024]. Lipschitz-guided design proposes the selection criterion
 132 tested [Chen et al., 2025]. Solver-specific schedules and exact linear parts are documented in
 133 diffusion practice [Sabour et al., 2024, Karras et al., 2022, Lu et al., 2022, Zhang and Chen, 2023].
 134 Stéphanovitch [2026] prove sharp Lipschitz regularity of canonical flow-matching fields and Euler-
 135 type W_2 rates; that is a bound, not a pairwise ranking of two interpolants at equal NFE.

136 Tao and Choi [2026] decompose $\nabla v = S + \Omega$ and control Euler error by the logarithmic norm
137 $\mu_2 = \lambda_{\max}(S)$, which is signed, whereas $\mathcal{R}$ integrates $\|A\|_2^2$. In one dimension $\mu_2(a) = a$ and
138 $\|A\|_2 = |a|$. Their Theorem 5 states that McCann displacement interpolation has vanishing material
139 derivative, hence Euler is order 2 in the Eulerian analysis; Remark 7 records the stronger Lagrangian
140 exactness observed on that interpolant. Neither statement applies to the independent-coupling linear
141 or VP path studied here. The present paper supplies a rationally certified Gaussian instance of ranking
142 failure for the unsigned average $\|A\|_2^2$. A counterexample to μ_2 itself is outside our scope. Benton
143 et al. [2024] bound W_2 of a learned flow; that is an error bound, not a pairwise ranking. Consistency
144 models [Song et al., 2023] distill few-step maps. Complementary solver-aware methods use temporal
145 information rather than an integrated spatial-Jacobian energy. Gupta et al. [2026] calibrate Euler
146 grids from trajectory acceleration on pretrained flow-matching models. The present contribution
147 is neither a sampler nor a training regularizer: it gives a certified ranking non-implication and a
148 solver-grid-aware impossibility result for the stated classes.

149 8 Limitations

150 The systems are commuting affine Gaussian interpolants, three fixed-step Runge–Kutta methods, and
151 small equal-NFE budgets. No learned neural field is evaluated. Chen et al. do not claim or prove
152 a discretization-error bound, nor that A_2 universally ranks equal-NFE solver error. Among four
153 specified candidate paths, the per-mode path wins 36 of 36 tested blocks; that does not imply that a
154 global $\mathcal{R}$ -minimizer minimizes fixed-NFE error. Theorem 1 is grid-aligned by construction. Cell
155 counts are hierarchical descriptive summaries, not population rates.

156 9 Conclusion

157 An integrated unsigned Jacobian-energy functional does not contain enough solver-specific temporal
158 information to universally determine few-step endpoint error. Averaged squared Jacobian regularity
159 can misrank equal-NFE Gaussian W_2 already for two C^1 interpolants of independent $N(0, 1)$ and
160 $N(0, 4)$. A grid-aware construction shows that even the global $\mathcal{R}$ -minimizer need not minimize
161 fixed-grid Euler error. A specified finite enumeration contains both agreement (36 of 36 tested blocks
162 among four candidate paths) and pairwise disagreement.

163 References

- 164 Michael S. Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying
165 framework for flows and diffusions. *Journal of Machine Learning Research*, 26(209):1–80, 2025.
- 166 Joe Benton, George Deligiannidis, and Arnaud Doucet. Error bounds for flow matching methods.
167 *Transactions on Machine Learning Research*, 2024. arXiv:2305.16860.
- 168 Yifan Chen, Eric Vanden-Eijnden, and Jiawei Xu. Lipschitz-guided design of interpolation schedules
169 in generative models. *arXiv preprint arXiv:2509.01629*, 2025. version 3, 16 May 2026.
- 170 Matthias Gelbrich. On a formula for the L^2 Wasserstein metric between measures on Euclidean and
171 Hilbert spaces. *Mathematische Nachrichten*, 147:185–203, 1990.
- 172 Aditi Gupta, Soon Hoe Lim, Annan Yu, and N. Benjamin Erichson. Sharpen your flow: Sharpness-
173 aware sampling for flow matching. *arXiv preprint arXiv:2605.11547*, 2026.
- 174 Ernst Hairer, Syvert P. Nørsett, and Gerhard Wanner. *Solving Ordinary Differential Equations I:
175 Nonstiff Problems*. Springer, 2nd edition, 1993.
- 176 Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-
177 based generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
178 arXiv:2206.00364.
- 179 Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow
180 matching for generative modeling. In *International Conference on Learning Representations
181 (ICLR)*, 2023. arXiv:2210.02747.

- 182 Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and
183 transfer data with rectified flow. In *International Conference on Learning Representations (ICLR)*,
184 2023. arXiv:2209.03003.
- 185 Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. DPM-solver: A fast
186 ODE solver for diffusion probabilistic model sampling in around 10 steps. In *Advances in Neural
187 Information Processing Systems (NeurIPS)*, 2022. arXiv:2206.00927.
- 188 Amirmojtaba Sabour, Sanja Fidler, and Karsten Kreis. Align your steps: Optimizing sampling
189 schedules in diffusion models. In *International Conference on Machine Learning (ICML)*, 2024.
190 arXiv:2404.14507.
- 191 Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben
192 Poole. Score-based generative modeling through stochastic differential equations. In *International
193 Conference on Learning Representations (ICLR)*, 2021. arXiv:2011.13456.
- 194 Yang Song, Prafulla Dhariwal, Mark Chen, and Ilya Sutskever. Consistency models. In *International
195 Conference on Machine Learning (ICML)*, 2023. arXiv:2303.01469.
- 196 Arthur Stéphanovitch. Lipschitz regularity in flow matching and diffusion models: Sharp sampling
197 rates and functional inequalities. *arXiv preprint arXiv:2604.06065*, 2026.
- 198 Chenxi Tao and Seung-Kyum Choi. On the role of strain and vorticity in numerical integration error
199 for flow matching. *arXiv preprint arXiv:2605.06680*, 2026.
- 200 Alexander Tong, Kilian Fatras, Nikolay Malkin, Guillaume Hugué, Yanlei Zhang, Jarrid Rector-
201 Brooks, Guy Wolf, and Yoshua Bengio. Improving and generalizing flow-based generative
202 models with minibatch optimal transport. *Transactions on Machine Learning Research*, 2024.
203 arXiv:2302.00482.
- 204 Qingsheng Zhang and Yongxin Chen. Fast sampling of diffusion models with exponential integrator.
205 In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2204.13902.

206 A Rational VP Heun certificate

207 This appendix records the integer comparison used in Proposition 1. It is part of this PDF, not a
208 separate supplement file.

209 The four-step VP Heun product is the following element $P(\pi, \sqrt{2})$ of $\mathbb{Q}[\pi, \sqrt{2}]$, obtained by
210 expanding the four Heun factors with $h = 1/4$ and reducing $\sqrt{2}^2 = 2$:

$$P(\pi, \sqrt{2}) = 1 + \frac{3}{40}\pi + \frac{15}{164}\pi\sqrt{2} + \frac{78579}{10758400}\pi^2 + \frac{45}{5248}\pi^2\sqrt{2} + \frac{11421}{17213440}\pi^3 + \frac{4671}{17213440}\pi^3\sqrt{2} \\ + \frac{5103}{275415040}\pi^4 + \frac{567}{55083008}\pi^4\sqrt{2} + \frac{243}{2203320320}\pi^5 + \frac{729}{2203320320}\pi^5\sqrt{2} + \frac{729}{176265625600}\pi^6. \quad (3)$$

211 Each of the twelve coefficients is a ratio of nonnegative integers, hence nonnegative, and the constant
212 term is 1. Substituting $\pi < 355/113$ and $\sqrt{2} < 99/70$ is therefore valid and yields

$$P\left(\frac{355}{113}, \frac{99}{70}\right) = \frac{192113353671412470139303}{102753427879939708813312} < \frac{187}{100},$$

213 equivalently

$$19211335367141247013930300 < 19214891013548725548089344.$$

214 Hence $r_{\text{VP}} < 187/100$ and $W_2^{\text{VP}} > 13/100$. The bound $\sqrt{2} < 99/70$ is $99^2 - 2 \cdot 70^2 = 1$;
215 $\pi < 355/113$ follows from Machin's formula with Leibniz remainders (seven Taylor terms for
216 $\arctan(1/5)$, four for $\arctan(1/239)$). An independent mpmath 1.3.0 interval enclosure is a
217 cross-check, not the proof.